

Data Quality Report

1. Overview

This report will look at the dataset of California wildfire accident. The initial dataset has 33 columns and 10,000 rows. All the workings are in the Jupyter Notebook, which is divided into 4 parts. Part 1 is about findings which are presented in this Data Quality Report. Part 2 is about creating data quality plan and display solutions to missing values. Part 3 is looking at different relationships between different columns and how they relate to each other. Part 4 is implementation of external sources to make the dataset better in quality.

This report will look closely at the Part 1 and initial findings from the original dataset. On the first sight, there was good amount of data, not many missing values, neither no duplicate columns were found. The only constant values were under the feature "State" and "Hazard Type". The main parts discussed were regarding values for continuous and categorical data. Part of this report, many logical tests were conducted to had better understanding of missing data. Also, in this dataset, "Damage" has only 2 unique values – "No Damage" or "Damage > 50%".

After the review of logical tests and how many columns are failing these tests, there will be drop of columns taken into account in the future analysis. In this dataset, from the continuous features, there were not many failing the tests. On the other hand, there are many columns regarding to location and these columns are not needed in the future analysis. For example, street name, street number, zip code, county and site address, they all refer to the same address information, which can be taken from latitude and longitude. There are very good columns that will be helpful in the future of this analysis. Building structures and different parts which can be found on the building, these features do not have many rows missing or unknown. This will be very helpful in determining the predictions for further questions.

2. Review logical integrity

23 logical tests were conducted to find out how many missing values is failing this dataset and will not be needed in the future analysis.

- Test 1: Multi-unit buildings should have more than 1 unit. The result: 0
- Test 2: distance from propane tank to the building being NaN values. Test result: 8000.
- Test 3: how many year-built parcel has 0 or string values. Test result: 3236.
- Test 4: any assessed values that are negative. Test result: 0
- Test 5: street number should not be null if street name is provided. Test result: 24
- Test 6: if the zip code format is within the standard US format. Test result: 1900 failing.
- Test 7: how many rows in structure category is NaN or empty. Test result: 0
- Test 8: hazard type should not be null if damage is reported. Test result: 0
- Test 9: latitude and longitude should be between -90 and 90; -180 and 180. Test result: 0
- Test 10: roof construction should not be null or empty. Test result: 353
- Test 11: Zip codes if they are presented if State is filled. Test result: 4064

These tests were tested if columns ".isnull":

- Distance residence to utility: 8527
- Window pane: 377
- Roof construction: 353
- Deck/porch on grade/elevated: 1632

- Structure type: 0
- Fence attached to building: 1632
- Damage: 0
- Call fire unit: 0
- Structure defence actions taken: 7276
- Structure category: 0
- Eaves: 391
- Vent screen: 397
- Exterior siding: 370
- Patio cover: 1633

3. Review continuous features

There were 9 features categorized as continuous features. The observations were following:

- a. % of Missing data = the most missing data had column units in structure with 70% of missing data, followed by year-built parcel with 25% and assessed value parcel with 5%. Otherwise, the rest of columns did not have any missing data.
- b. Cardinality = cardinality was very high and good for this dataset. Longitude and latitude had 9995 unique values, which is almost perfect score. Assessed value parcel around 7500 unique values and year-built parcel 140 unique values, these indicate a solid number for cardinality for this analysis. On the other hand, units in structure were 15, which is very good, but because of 70% of missing values, this column will be dropped down in the future analysis.
- c. Further data in the table = year-built parcel has min value 0, which indicates that this will have to be improved in the future analysis. The mean (50%) is 1962, which looks like a good indicator. Assessed improved value has minimum as 0 as well, but the quartiles look good overall, therefore no major improvements will be made. The quartiles are steadily increasing, which is showing that the data is in a good shape. Units in structure have many 0 among the table, therefore these data are not indicating anything.

4. Review categorical features

There were 23 features categorized as categorical features. From the data table, there were found these observations.

- a. % of Missing Data = structure defence actions taken, zip code and community had the most missing data. Therefore, they were dropped down in the further analysis. Street type, city, deck/porch on grade/elevated, patio attached to structure, and fence attached, these columns had only around 15% of missing data, therefore it would not be efficient to drop these. The rest of columns had 0-5% of missing data, therefore there were not remade in any way.
- b. Cardinality = street number, street name, site address parcel had very high cardinality, unfortunately there was still plenty of missing data. Roof construction, eaves, vent screen, exterior siding, windowpane, deck/porch on grade/elevated, patio attached to structure, fence attached to structure had around 6-10 unique values, which indicates a good range of different building structures for further analysis. City, county, community, and street type had a very poor unique values –

these columns are expected to have cardinality of 100%, however they only had around 400 unique values. Damage, state and hazard type had the expected cardinality of 1-2, this is good as they can be dropped down from the analysis and no false assumptions will be made.

- c. Top features = in the categorical table, it is shown that slightly above 50% the damage reported in this dataset was “Destroyed > 50%”, with this connected it is also, that most destroyed structures were a type of single residence, with no porch/deck, and no fence. The most reported destroyed county was Los Angeles.

5. Action to take – briefly the actions to take

4 main actions are going to be taken in the analysis:

- Dropping values that have many missing data, and they are not appearing as clean data.
- Replacing categorical columns that have NaN values with their mode
- Replacing missing values in continuous columns with their median
- In the rest of columns, fill in the missing values with the word “Unknown”

Appendix – Tables, Graphs, List of Features

List of Categorical and Continuous Features

• _Damage	category
• _Street_Number	category
• _Street_Name	category
• _Street_Type_eg_road_drive_lane_etc	category
• _City	category
• State	category
• Zip_Code	category
• _CAL_FIRE_Unit	category
• County	category
• Community	category
• Incident_Start_Date	category
• Hazard_Type	category
• Structure_Defense_Actions_Taken	category
• _Structure_Type	category
• Structure_Category	category
• _Units_in_Structure_if_multi_unit	continuous
• _Roof_Construction	category
• _Eaves	category
• _Vent_Screen	category
• _Exterior_Siding	category
• _Window_Pane	category
• _DeckPorch_On_Grade	category
• _DeckPorch_Elevated	category
• _Patio_CoverCarport_Attached_to_Structure	category
• _Fence_Attached_to_Structure	category
• Distance__Propane_Tank_to_Structure	category
• Distance__Residence_to_UtilityMisc_Structure_gt_120_SQFT	category
• Assessed_Improved_Value_parcel	continuous
• Year_Built_parcel	continuous
• Site_Address_parcel	category
• Latitude	continuous
• Longitude	continuous

Continuous features table

	count	mean	std	min	25%	50%	75%	max
_Units_in_Structure_if_multi_unit	3024.0	0.301	2.016e+00	0.00	0.000	0.000	0.000	8.000e+01
Assessed_Improved_Value_parcel	9477.0	691595.909	6.358e+06	0.00	69151.000	159866.000	342085.000	3.873e+08
Year_Built_parcel	7563.0	1757.238	6.049e+02	0.00	1940.000	1962.000	1983.000	2.022e+03
Latitude	10000.0	37.379	2.490e+00	32.60	34.196	38.476	39.744	4.195e+01
Longitude	10000.0	-120.539	1.823e+00	-123.47	-122.139	-121.455	-118.539	-1.164e+02

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	%missing
OBJECTID	0.00
_Units_in_Structure_if_multi_unit	69.76
Assessed_Improved_Value_parcel	5.23
Year_Built_parcel	24.37
Latitude	0.00
Longitude	0.00

Feature	Cardinality
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OBJECTID	10000
_Units_in_Structure_if_multi_unit	15
Assessed_Improved_Value_parcel	7618
Year_Built_parcel	141
Latitude	9993
Longitude	9996

Categorical features table

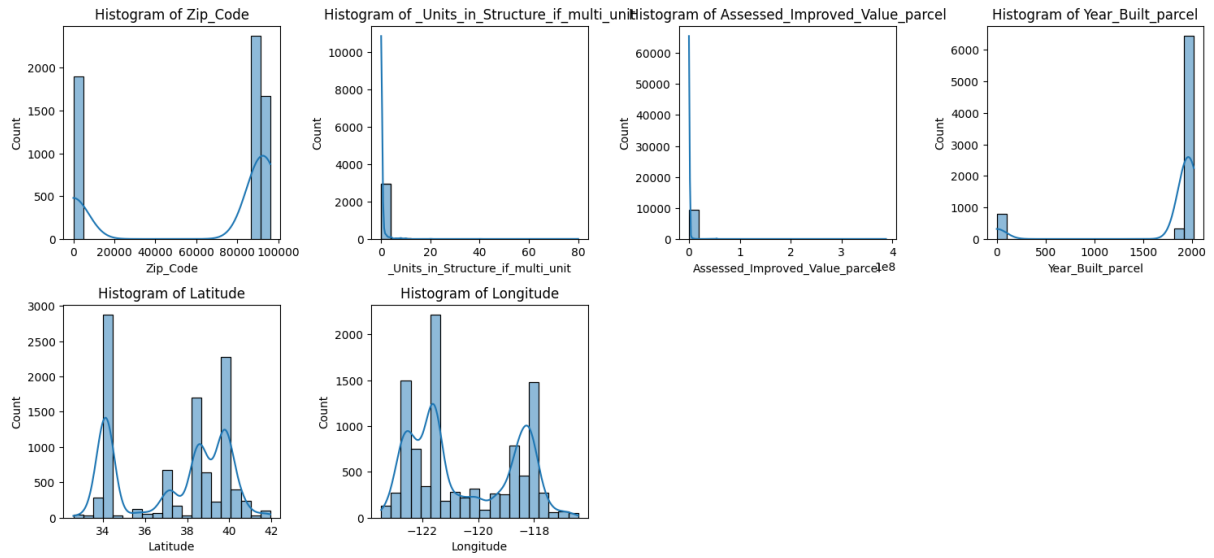
Percentage of Missing Values:

	perc_missing
OBJECTID	0.00
_Damage	0.00
_Street_Number	3.30
_Street_Name	4.06
_Street_Type_eg_road_drive_lane_etc	10.77
_City	24.48
State	0.01
Zip_Code	40.65
_CAL_FIRE_Unit	0.00
County	0.02
Community	57.70
Hazard_Type	0.00
Structure_Defense_Actions_Taken	72.76
Structure_Category	0.00
_Roof_Construction	3.53
_Eaves	3.91
_Vent_Screen	3.97
_Exterior_Siding	3.70
_Window_Pane	3.77
_DeckPorch_On_Grade	16.32
_DeckPorch_Elevated	16.32
_Patio_CoverCarport_Attached_to_Structure	16.32
_Fence_Attached_to_Structure	16.32
Site_Address_parcel	5.87

Cardinality Table:

	card
OBJECTID	10000
_Damage	2
_Street_Number	5684
_Street_Name	4010
_Street_Type_eg_road_drive_lane_etc	23
_City	250
State	1
Zip_Code	148
_CAL_FIRE_Unit	27
County	45
Community	420
Hazard_Type	1
Structure_Defense_Actions_Taken	10
Structure_Category	6
_Roof_Construction	9
_Eaves	5
_Vent_Screen	6
_Exterior_Siding	10
_Window_Pane	4
_DeckPorch_On_Grade	5
_DeckPorch_Elevated	5
_Patio_CoverCarport_Attached_to_Structure	4
_Fence_Attached_to_Structure	4
Site_Address_parcel	8499

Graphs for continuous features



Graphs for categorical features

